

Complex integration

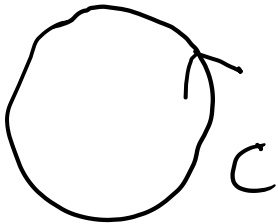
Line integral in complex plane

$$\int_C f(z) dz$$

$z = x + iy$

$\rightarrow x/y/\text{parameter } t$

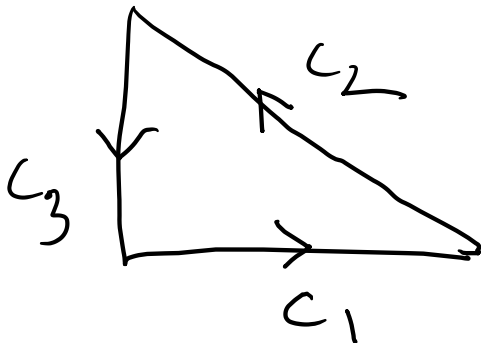
$$\int f(x) dx / \int f(y) dy / \int f(t) dt$$



$$\oint_C f(z) dz$$



$$C = C_1 + C_2 + C_3$$



$$\oint_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz + \int_{C_3} f(z) dz$$

1. Evaluate $\int_C \bar{z} dz$, where C is given by $x = 3t, y = t^2; -1 \leq t \leq 4$

$$\Rightarrow z = x + iy$$

$$\Rightarrow z = 3t + it^2$$

$$\therefore dz = (3 + 2it) dt$$

$$\bar{z} = 3t - it^2$$

$$\int_C \bar{z} dz = \int_{-1}^4 (3t - it^2) \cdot (3 + 2it) dt$$

$$= \int_{-1}^4 (9t + 6it^2 - 3it^2 + 2t^3) dt$$

$$= \int_{-1}^4 (9t + 2t^3 + i3t^2) dt$$

$$= \int_{-1}^4 (9t + 2t^3) dt + i \int_{-1}^4 3t^2 dt$$

$$= 195 + 65i$$

$$x = 3t + it^2 \checkmark$$

$$\frac{dz}{dt} = \frac{d}{dt} (3t + it^2)$$

$$\Rightarrow \frac{dz}{dt} = 3 + i2t$$

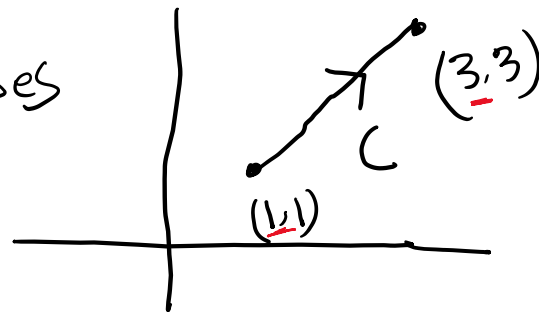
$$\Rightarrow dz = (3 + 2it) dt$$

$$-2it^3$$

(1,1) (3,3)

2. Sketch the path C from $z = 1 + i$ to $z = 3 + 3i$ and hence evaluate $\int_C \operatorname{Re} z \, dz$.

⇒ Eqn of straight line which passes through (1,1) & (3,3) is



$$\frac{x-1}{1-3} = \frac{y-1}{1-3}$$

$$\Rightarrow x-1 = y-1$$

$$\Rightarrow x = y$$

$$\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2}$$

$$z = 3x$$

$$dz = 3 \, dx$$

$$z = x + iy$$

$$z = x + ix \quad \therefore \operatorname{Re}(z) = x$$

$$z = (1+i)x$$

$$dz = (1+i) \, dx$$

$$z = x + iy$$

$$= y + iy$$

$$= (1+i)y$$

$$\int_C \operatorname{Re} z \, dz = \int_1^3 x(1+i) \, dx$$

$$\begin{aligned} &= (1+i) \int_1^3 x \, dx = (1+i) \left[\frac{x^2}{2} \right]_1^3 \\ &= 4(1+i) \end{aligned}$$

$$\int \ln x = x \ln x - x$$

3. $\int_C \ln(z) dz$, C is the shortest path from i to $2i$. ^(0,1) ^(0,2)

$\Rightarrow$ The eqⁿ of straight line is $x=0$

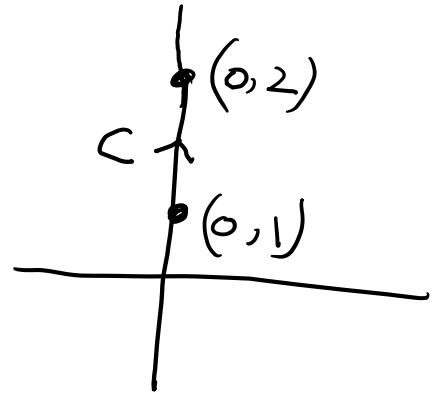
$$z = x + iy$$

$$z = iy$$

$$dz = i dy$$

$$\begin{aligned} \int_C \ln z \, dz &= \int_1^2 \ln(iy) \, i dy \\ &= \int_i^{2i} \ln u \, du \\ &= [u \ln u - u]_i^{2i} \end{aligned}$$

$$\begin{aligned} &= (2i \ln(2i) - 2i) - (i \ln i - i) \\ &= 2i \ln(2i) - i \ln i - i \end{aligned}$$



$$\int \ln(2x) \, dx$$

$u = 2x$

$$u = iy$$

$$du = i dy$$

y	1	2
u	i	$2i$

4. Sketch the path C consisting of two line segments, one from $z = 0$ to $z = 2$ and other from $z = 2$ to $z = 3+i$, hence evaluate $\int_C z^2 dz$.

$\Rightarrow$ Along C_1 :

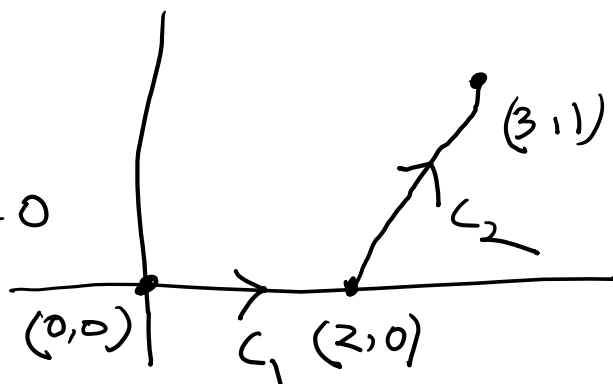
The eqⁿ of Straight line is $y=0$

$$z = x + iy$$

$$z = x$$

$$dz = dx$$

$$\int_{C_1} z^2 dz = \int_0^2 x^2 dx = \left[\frac{x^3}{3} \right]_0^2 = \frac{8}{3}$$



Along C_2 : Eqⁿ of Straight line is

$$\frac{x-2}{2-3} = \frac{y-0}{0-1}$$

$$\Rightarrow x-2 = y$$

$$\Rightarrow x = 2 + y$$

$$z = x + iy$$

$$= 2 + y + iy$$

$$z = 2 + (i+1)y \quad \therefore dz = (i+1)dy$$

$$\int_{\gamma_2} z^\vee dz = \int_0^1 (z + (i+1)y)^\vee \cdot (i+1) dy$$

$$= \int_2^{3+i} u^\vee du$$

$$= \left[\frac{u^3}{3} \right]_2^{3+i}$$

$$= \frac{10}{3} + \frac{26}{3}i$$

$$u = z + (i+1)y$$

$$du = (i+1)dy$$

y	0	1
u	2	3+i

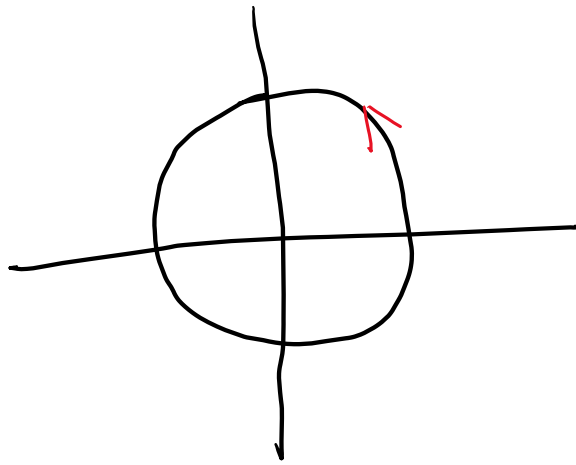
$$\frac{(3+i)^3}{3} - \frac{2^3}{3}$$

$$\therefore \int_C f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz$$

$$= \frac{8}{3} + \frac{10}{3} + \frac{26}{3}i$$

$$= 6 + \frac{26}{3}i$$

$$|z|=1$$

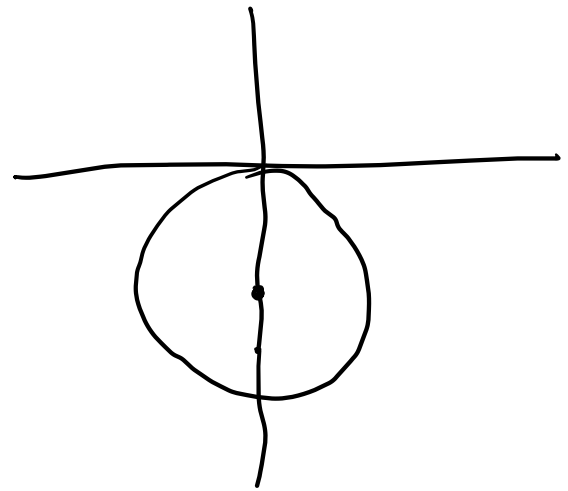


$$|z+3i|=3$$

$$\Rightarrow |x+iy+3i|=3$$

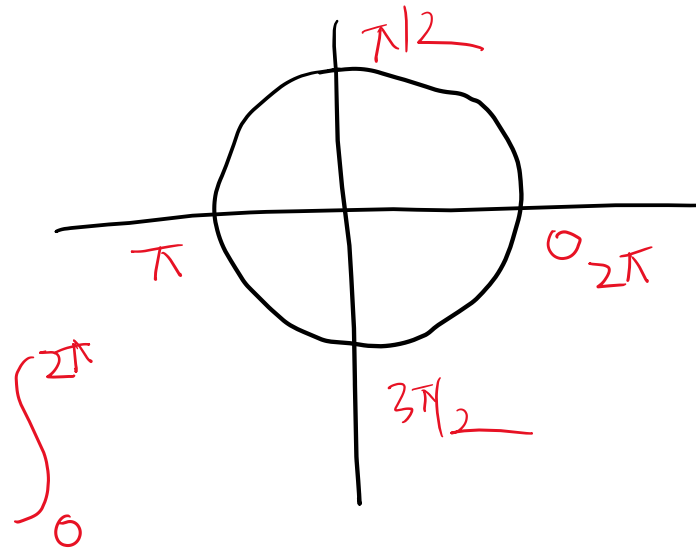
$$\Rightarrow |x+i(y+3)|=3$$

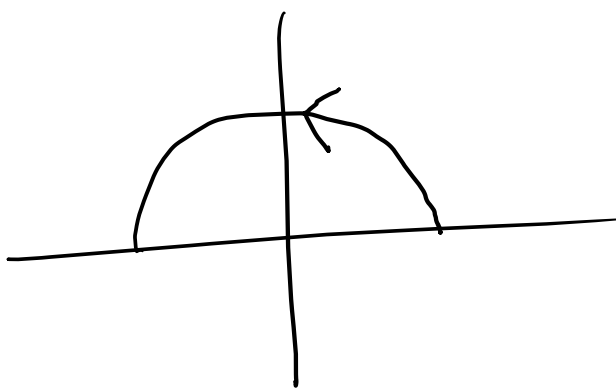
$$\Rightarrow x^2 + (y+3)^2 = 3$$



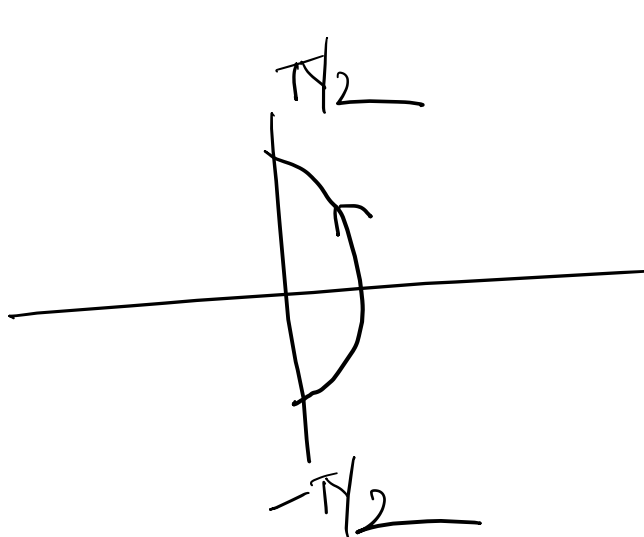
$$z = re^{i\theta}$$

$$dz = r i e^{i\theta} d\theta$$





$$\int_0^{\pi}$$

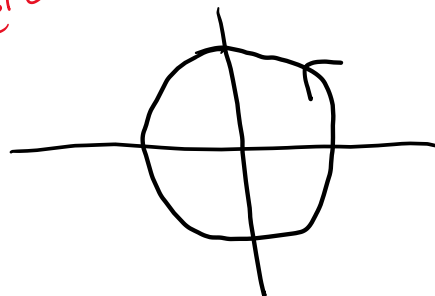


$$\int_{-\pi/2}^{\pi/2}$$

$$2 \times \frac{1}{2}$$

5. Evaluate $\int_C (z + z^{-1}) dz$ along the path $C, |z| = 2$.

$$\frac{1}{2e^{i\theta}}$$



$$\Rightarrow |z| = 2$$

$$z = 2e^{i\theta} \quad z^{-1} = \frac{1}{2}e^{-i\theta}$$

$$dz = 2ie^{i\theta} d\theta$$

$$\begin{aligned} \int_C (z + z^{-1}) dz &= \int_0^{2\pi} \left(2e^{i\theta} + \frac{1}{2}e^{-i\theta} \right) 2ie^{i\theta} d\theta \\ &= \int_0^{2\pi} (4ie^{2i\theta} + i) d\theta \end{aligned}$$

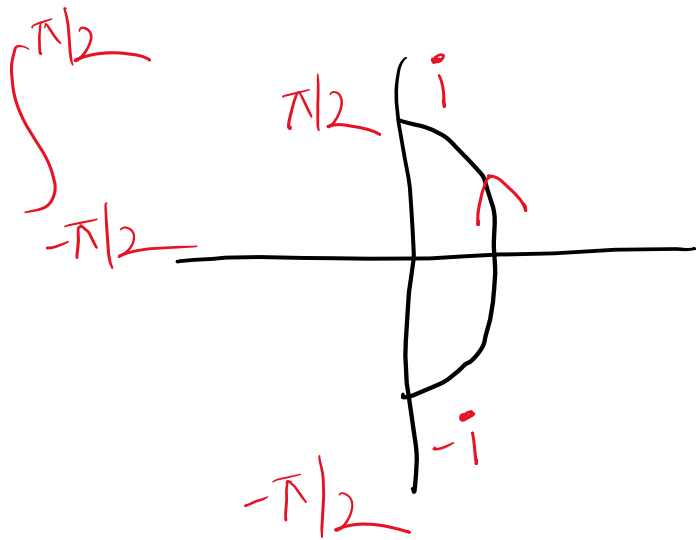
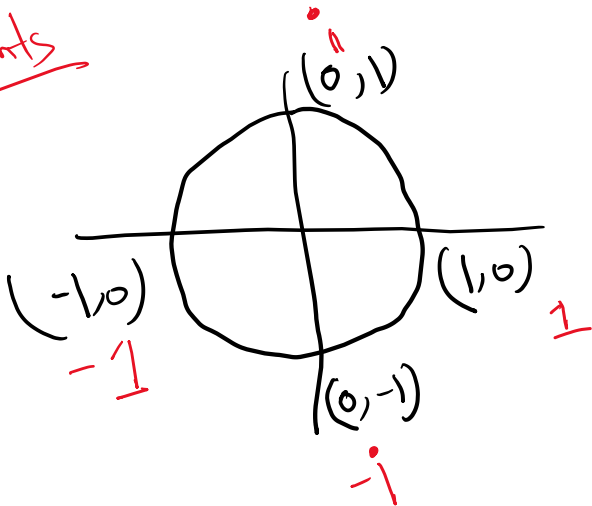
$$\begin{aligned}
 &= \left[4i \frac{e^{2i\theta}}{2i} + i\theta \right]_0^{2\pi} \\
 &= \left[2e^{i2\theta} + i\theta \right]_0^{2\pi} \\
 &= 2e^{i4\pi} + i2\theta - 2
 \end{aligned}$$

H.W $\rightarrow 1 + \frac{1}{z}$

6. $\int_C \frac{z+1}{z} dz$, where C is the right half of the circle $|z| = 1$ from $z = -i$ to $z = i$

$2i + i\pi$

Hints



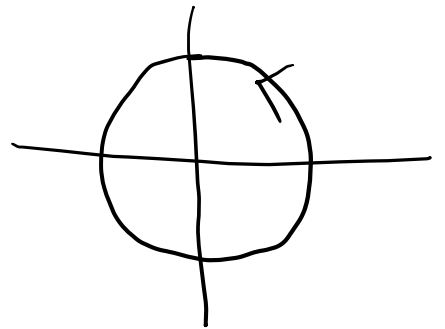
7. Evaluate $\int_C \operatorname{Re} z \, dz$ along the path C , which is the circle $|z| = 1$.

$$|z| = 1$$

$$\Rightarrow z = e^{i\theta} \rightarrow \operatorname{Re}$$

$$\Rightarrow z = \cos\theta + i\sin\theta \quad \therefore \operatorname{Re}(z) = \cos\theta$$

$$dz = (-\sin\theta + i\cos\theta) d\theta$$



$$\int_C f(z) dz = \int_0^{2\pi} (\cos \theta) (-\sin \theta + i \cos \theta) d\theta$$

$$= \int_0^{2\pi} -\sin \theta \cos \theta d\theta + i \int_0^{2\pi} \cos^2 \theta d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} 2 \sin \theta \cos \theta d\theta + \frac{i}{2} \int_0^{2\pi} 2 \cos^2 \theta d\theta$$

$$= 0 + i\pi$$

$$= i\pi$$

$$= -\frac{1}{2} \int_0^{2\pi} \sin 2\theta d\theta + \frac{i}{2} \int_0^{2\pi} (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{2} \left[\frac{\cos 2\theta}{2} \right]_0^{2\pi} + \frac{i}{2} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{2\pi}$$

$$= \frac{1}{4} (1-1) + \frac{i}{2} [(2\pi+0) - 0]$$

$$= 0 + i\pi$$

$$= i\pi$$